CALIBER 2026 | BUSINESS PROPOSAL | PRELIMINARY STAGE

Asset Knowledge Platform

A manufacturing knowledge hub for the LLDPE unit: one governed database behind a Gemini-powered assistant, with live P&ID, OPL simulation, failure memory and predictive maintenance on the same asset tags.

Chosen use case: Case 1, Manufacturing Knowledge Hub

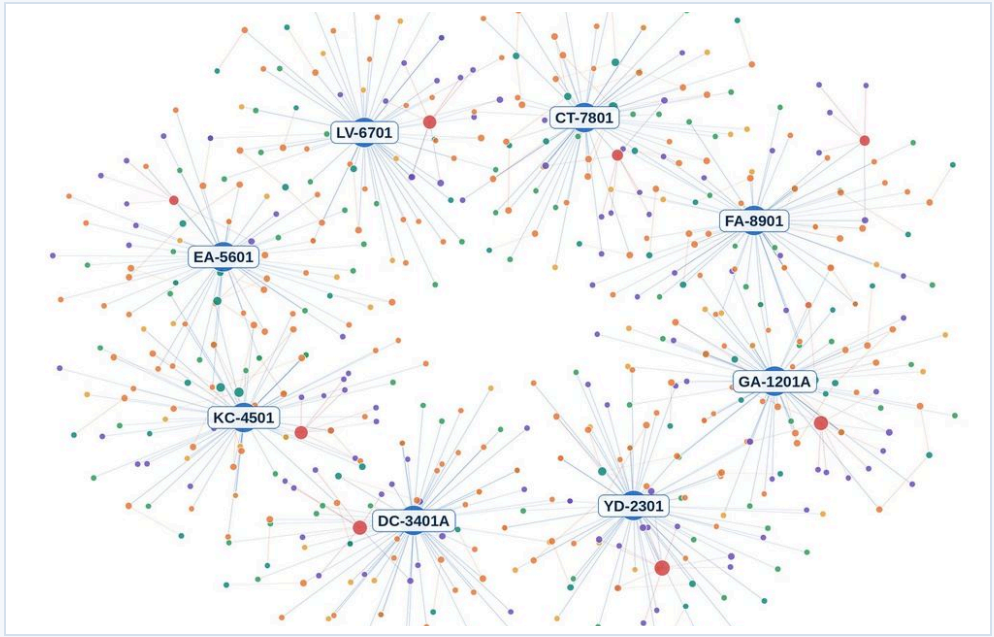
564
linked records

1,109
searchable passages

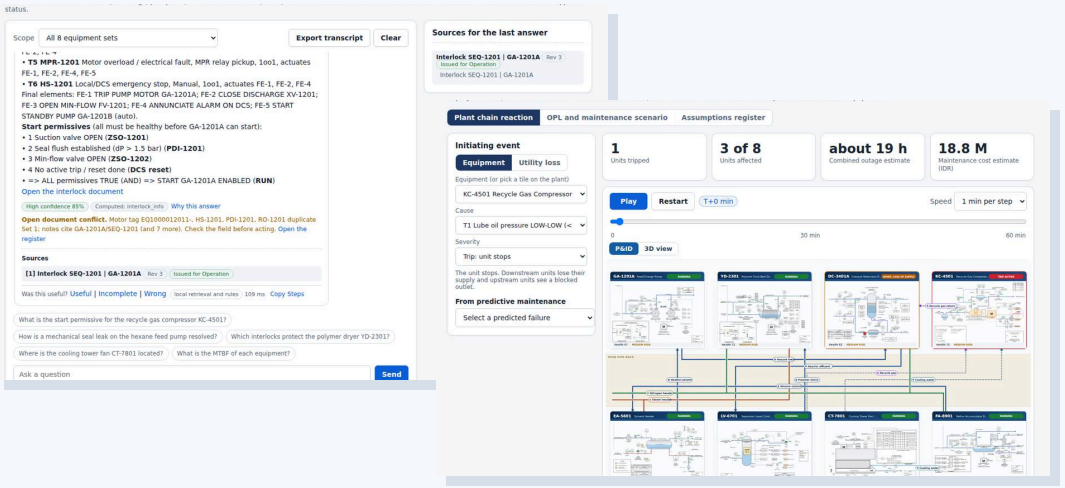
20
computed AI tools

128/128
evaluation questions passed

Team: [Team name]
University: [University] **Supervisor:** [Supervisor name]



Knowledge network of the case data: 8 equipment tags and the 556 records that hang on them

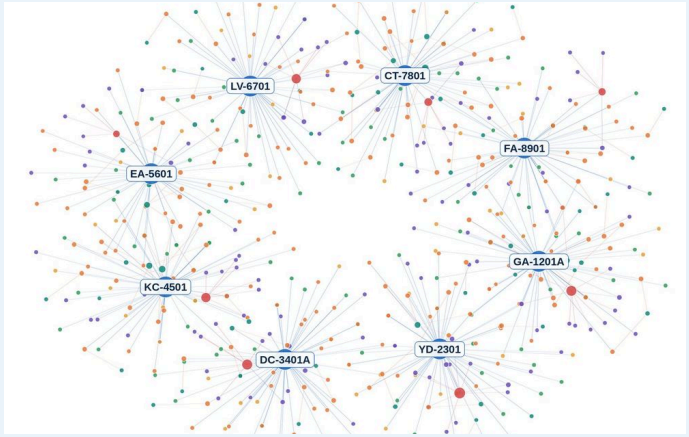


Working prototype on the case baseline data, with a 3-minute demonstration video

Problem Analysis: the Knowledge Gap Between What the Plant Records and What Engineers Can Use at the Point of Work

CALIBER 2026
Case 1

Knowledge landscape of the baseline data



Equipment 8 · Documents 96 · Work orders 211 · Instruments 84 · OPL 56 · Components 19 · Interlocks 8

Analysis

564 records of 7 types are joined only by the equipment tag. No source system stores those links; the platform builds 624 of them.

Key finding

Reading the sources together exposes 48 conflicts, for example valve FV-2302 labelled both trips OPEN and trips CLOSED.

Tension

211 work orders and 63 failures, none linked to the lesson that applies. Only 21% of faults found before turning corrective.

Operating context

Digital agenda
Industry 4.0 award and a digital twin at Cilegon (public sources)

Safety-critical
Continuous, asset-intensive petrochemical operation

Seven functions
Production, Maintenance, Reliability, Warehouse, Procurement, Energy, HSE

Core problem: improper execution

Engineers need a correct step, a current drawing or a past lesson during start-up, troubleshooting and maintenance. The answer exists somewhere, but cannot be found fast, cannot be verified, and is not tied to the asset in front of them.

Key challenges

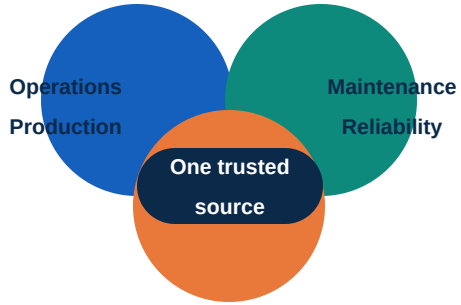
Scattered sources
Separate systems, no shared tag

Missing tacit knowledge
Why a trip occurs stays with senior staff

Conflicting documents
Revision and approval not visible at use

Disconnected live data
Work orders, trips, drawings unlinked

Stakeholder alignment



Needs

The exact step and set point at the valve

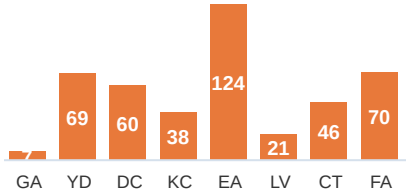
History, lessons, spares

Limits, hazards, permits

Audit and revision trail

Baseline impact

Breakdown downtime by unit, hours



Reading: EA-5601 alone lost 124 h; the same fault is investigated again.

31
breakdowns

434 h
downtime

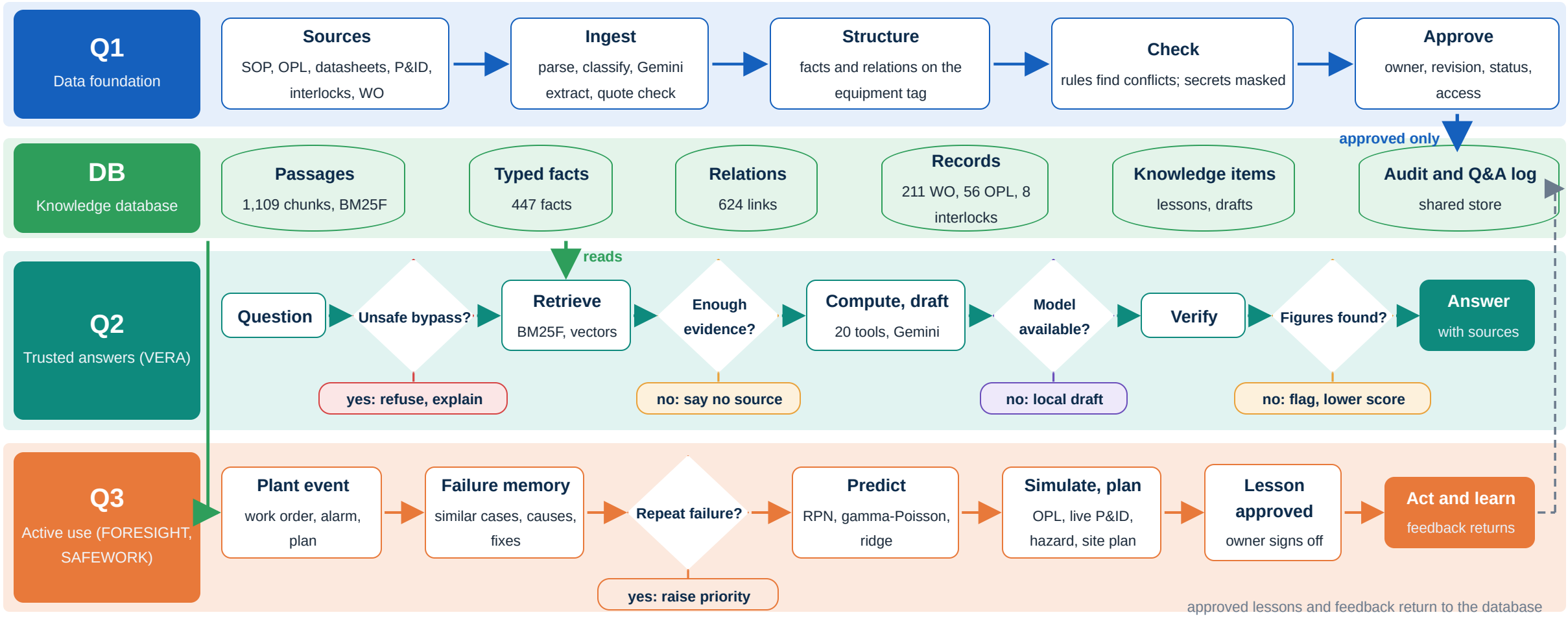
IDR 538 M
maintenance, 82% corrective

Conclusion

The plant already records what engineers need, but no system joins it to the asset at the point of work. That gap, not missing data, is what causes improper execution and repeat failures.

Source: CALIBER 2026 case book (Case 1); baseline data analysed in the prototype: 8 equipment sets, LLDPE unit, June 2024 to December 2025

Solution: One Governed Knowledge Database Fed by Data Operations, Read by Three Engines That Answer, Predict and Plan



Conclusion

Every answer and plan is read from one governed database, so a source is checked before it is used and a lesson is approved before it is reused. The free-tier model only words the answer; the numbers, quotes and safety refusals come from the platform.

Source: Prototype architecture and counts from the running application; Gemini 2.5 Flash is reached through the serverless proxy (free tier)

Business Impact: Each Feature Moves a Measured KPI, Worth IDR 1.5 B a Year in the Base Case With a 21-Month Payback

Platform capability	KPI and where the number comes from	Simple calculation	IDR M
AI assistant engineer finds the procedure, drawing and past case in minutes	**Search and verify time per engineer: 40 to 20 min a day** **Assumption:** pilot logs replace it	45 users x 20 min x 230 days / 60 = 3,450 h saved; x IDR 175,000 per h = IDR 604 M	604
Failure memory, predictive checks known causes and fixes shown at the fault; repeat jobs flagged	**MTTR 14.0 h to 12.6 h (10% less)** **Case data:** 211 WO; 10% assumed	20.4 breakdowns a year (31 in 1.52 yr) x 14.0 h x 10% = 28.6 h; x IDR 30 M per downtime hour = IDR 857 M	857
Knowledge Studio, OPL lessons new engineers learn from approved lessons and the simulator	**New engineers productive one month sooner** **Assumption:** HR data replaces it	One month of reduced output for new joiners, set at IDR 60 M a year	60
Interlock guard, readiness, hazard view	Improper executions: counted in the pilot	Not valued: no case data on cost per incident	not valued
Quality register, approvals	48 conflicts owned, 90% closed	Not valued: effect shows through the levers above	not valued

Result

IDR 1,521 M

a year in the base case: 604 + 857 + 60

IRR 40%

NPV at 10%: IDR 0.91 B

Net over 3 years: IDR 1.44 B

Payback 21 months

Low IDR 0.8 B, no payback

High IDR 2.1 B, 16 months

Payback: cost is IDR 1,400 M one-off and IDR 370 M a year. Year 1 net = 60% x 1,521 - 370 = 542; years 2 and 3 net = 1,521 - 370 = 1,151 each. After year 1 the gap is 1,400 - 542 = 858, and 858 / 1,151 x 12 = 9 months, so payback is at month 21.

Conclusion

Value rests on two measurable levers, search time and repair time, and every input is shown so the Company can replace it with its own number. The base case pays back in 21 months; the low case does not, and the pilot is designed to find out which case is real.

Source: Case data: 211 work orders, 31 breakdowns over 1.52 years, MTTR 14.0 h. Assumptions (grey): users, minutes saved, rates, 10% repair-time reduction; Company figures replace them in the pilot

Feasibility and Roadmap: Read-Only, Advisory Deployment in Four Phases With Explicit Decision Gates

Feasibility: six readiness dimensions

Dimension	Status	In place today	Still needed
Technical	Ready	Prototype end to end; SQL schema and API spec; replaceable model contract	Production database, ingestion queue
Operational	Ready	Advisory only; local search if the model is offline; fits OPL and work-order routines	Shift-team trial, champions per function
Legal	Decision needed	Drawing and vendor IP respected; data held in region	Enterprise AI terms with no training on Company data; PDP Law review
Cybersecurity	Planned	Roles, audit trail, secret masking, prompt-injection guards, rate limits	SSO, key vault, hub outside the OT zone (IEC 62443), red-team test
Data governance	Ready	Owner, revision and status on every record; 48 conflicts in a managed register	Named data stewards, refresh rules per source
Organisation	Planned	Roles and approval workflow built in	Two-hour role-based training, monthly KPI review, sponsor

Deployment path and budget

Prototype today	Static page, Netlify functions, Blobs, Gemini 2.5 Flash free tier, local fallback
Pilot (90 days)	Same stack; read-only EDMS and AIMS adapters, single sign-on, 45 users
Production	PostgreSQL with pgvector, ingestion queue, Company-approved model, SIEM export

One-off budget IDR 1,400 M; running cost IDR 370 M a year

Integration 650 | Data 350 | Change 250 | Sec. 150

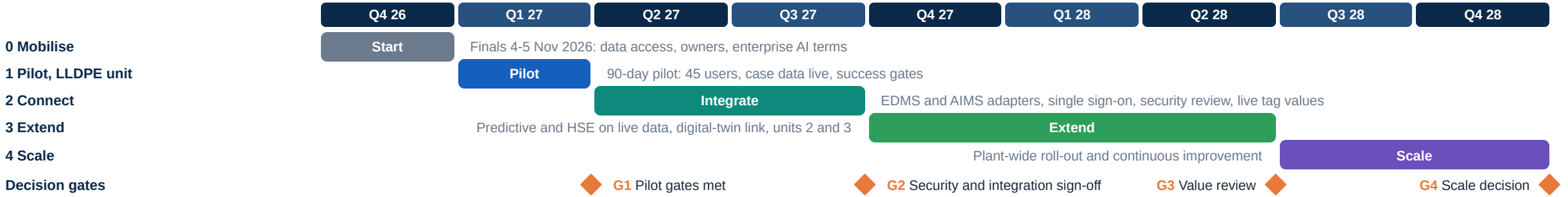
Wrong answer: evidence-only drafting, verification, refusals

Poor data quality: conflict register, named owners

Low adoption: champions, tools used in daily work

Model cost or outage: replaceable model, offline fallback

Implementation roadmap



Conclusion

The pilot is feasible today because the prototype already runs end to end, reads from Company systems only, and keeps a person in every approval. Four phases with decision gates keep spend at IDR 1.4 B one-off and let the Company stop at any gate.

Source: Prototype and architecture notes; dates and budget are proposals for discussion with the Company

Conclusion: A Trusted, Traceable Knowledge Hub Turns Scattered Plant Knowledge Into Faster, Safer and More Reliable Decisions

Summary of the proposal, how it answers the three key questions of the case, and the decision requested

CONNECT TAGLINK: one governed base

- 98 documents and 211 work orders unified on the equipment tag
- 48 conflicts detected and assigned to owners
- Ingestion, quality gates and approval for every record

98 to 1 documents into one governed base

ASK VERA: answers with evidence

- Source, revision, approval status and confidence on every answer
- Numbers and quotes verified against the cited text
- Safety bypass requests refused with the reason

128/128 evaluation questions passed

ACT FORESIGHT, SAFEWORK

- Failure memory with RCA and lessons at the point of fault
- Risk ranking, readiness, hazard screening and permit-aware planning
- Read-only links to EDMS, AIMS and the digital twin

20 tools computed from plant records

How the proposal answers the three key questions of the case

Q1	**Build a structured Industrial Data Ops foundation**	Tag-keyed data model over seven source types; AI-assisted ingestion, quality gates and approval with named owners; conflicts surfaced on day one	**Evidence:** 98 documents, 1,109 passages, 48 conflicts registered
Q2	**Help engineers find trusted answers faster and reduce improper execution**	Hybrid retrieval, evidence-weighted confidence score, verification of numbers and quotes, refusal of unsafe requests	**Evidence:** 128/128 questions passed; 20 min a day saved per engineer (assumption, slide 4)
Q3	**Integrate with operational systems for reliability and improvement**	Read-only EDMS, AIMS, historian and twin links; failure memory, backtested reliability analytics, readiness, hazard screening and site planning	**Evidence:** 64-forecast backtest; 20 computed tools

Decision requested: a 90-day pilot

Weeks 1 to 2
Agree data access, owners and AI terms

Weeks 3 to 6
Load the LLDPE unit and train 45 users

Weeks 7 to 12
Measure against the success gates, then decide on scale

Conclusion

The proposal gives engineers one trusted place to ask, with sources they can check, and turns each repair into a lesson the next shift can use. We ask for a 90-day pilot so the Company can measure the value before it commits to scale.

Source: Prototype test results (128 of 128 evaluation questions, static, regression and layout checks) and the business case on slide 4

Team Profile

Team name, university, supervisor and members (replace the bracketed fields before submission)

Team name: [Team name] **University:** [University] **Supervisor:** [Supervisor name]

No.	Name	Major	Semester	Area of expertise	Contribution to the proposal
1	[Name]	[Computer Science]	[6]	[AI and software]	[Formulated the idea; built the prototype and the AI assistant]
2	[Name]	[Chemical Engineering]	[7]	[Process and reliability]	[Problem analysis, process content and hazard models]
3	[Name]	[Industrial Engineering]	[7]	[Maintenance and business case]	[Business impact, KPIs and roadmap]

The booklet allows two to three members from the same university. Delete unused rows.

A1 The Database: One Chunk-and-Fact Model in the Browser, a Shared Store Today, PostgreSQL With pgvector as the Target

Every record becomes a chunk with a stable id, a status, a trust weight and an access list; numbers become typed facts; links become relations

Browser

akp_kb in IndexedDB (localStorage fallback)

- In-memory chunk list and term index, rebuilt when data changes
- Knowledge items, audit trail, feedback, answer log, evaluation runs, embeddings cache
- Works offline as a plain file

Shared store

Netlify Blobs through /api/kb

- Collections: uk, wo, fb, audit, qa
- Last write wins; a base version in the request turns a newer server copy into a conflict
- Tombstones propagate deletes; optional access codes; polling for changes

Production target

PostgreSQL 15 with pgvector and pg_trgm

- 25 tables, generated tsvector with GIN index, HNSW cosine index on vector(1536)
- Status vocabulary with trust weights, versions with validity dates
- Row-level role filter through acl_roles

Sync: the page pulls the shared revision, merges per item by timestamp and pushes local edits. If /api/health is unreachable the app keeps working from browser storage.

Assets and documents

asset	document	chunk	fact
component	document_version		relation

Operations

interlock	instrument	opl	opl_case
interlock_trigger	work_order	failure_event	

Knowledge and governance

knowledge_item	knowledge_review	qa_log	feedback
knowledge_revision		qa_citation	eval_case

Also: eval_run, audit_log, app_user, status_vocab (11 trust codes). Keys: chunk_id = kind:record#ordinal, tag is the common join key

Anatomy of one chunk

chunk_id	il:TJC-LLD-IL-KC-4501#03
kind / record	il / interlock sheet
title, section	Interlock | KC-4501 | T3
text	VSHH-4505 80 um, 2oo3 ...
tags	{KC-4501, VSHH-4505}
status, trust	approved, 1.00
revision, date	Rev 3, 2025-02-11
acl roles	all roles
hash, tokens	cyrb53, 112
embedding	vector(1536) when connected

Indexes: GIN on the generated tsvector and on tags; B-tree on asset and status; HNSW (cosine) on the embedding. Embeddings are recomputed only when the content hash changes.

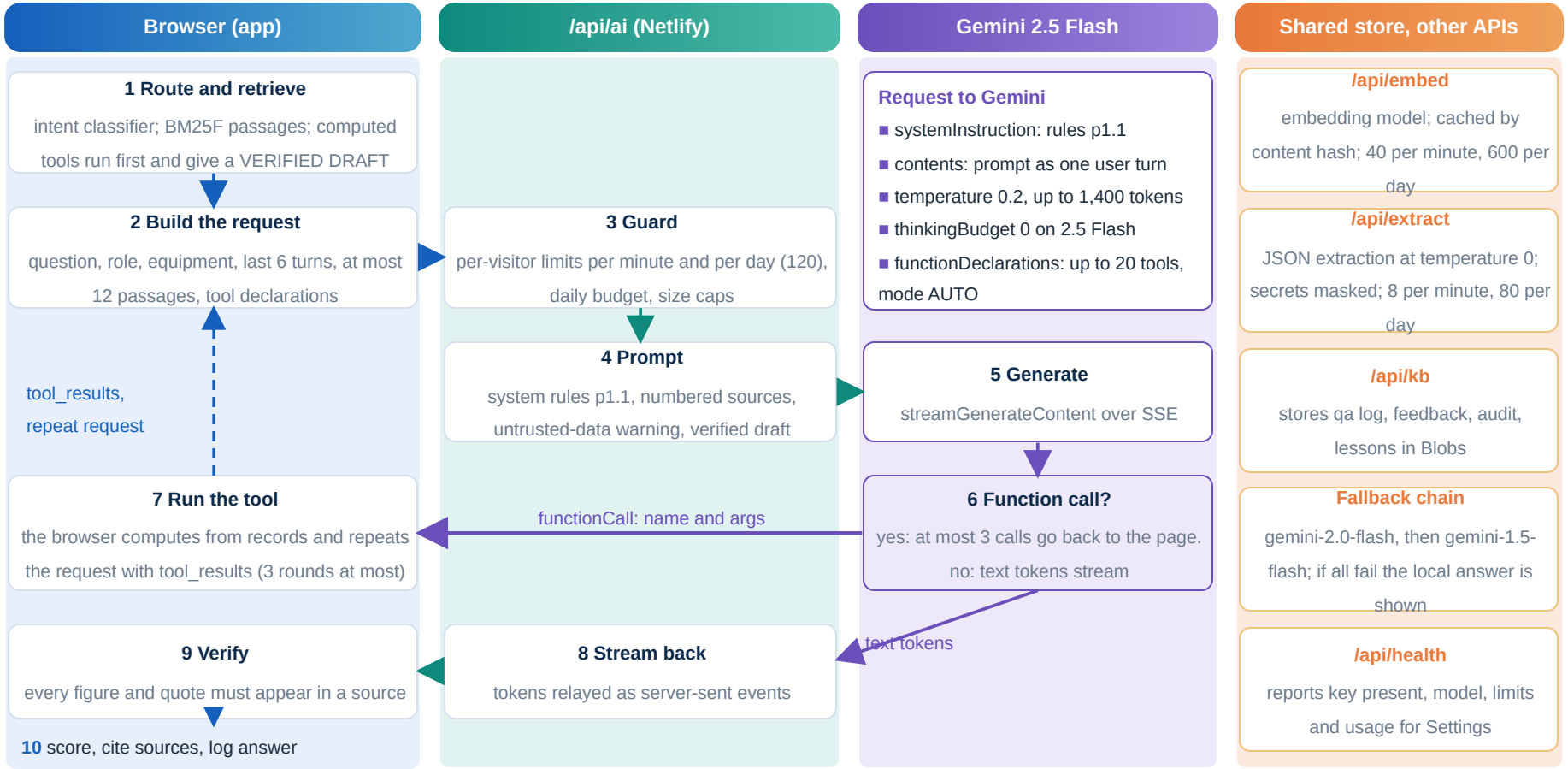
1,109	447	624	564	98	211
chunks	typed facts	relations	linked records	documents	work orders

Counts are read live from the prototype; the same shapes map one-to-one onto the SQL tables.

Source: Prototype data layer (dl.js, facts.js, sync.js), Netlify Blobs functions (kb.mjs) and the SQL schema in docs/DATA_MODEL.sql

A2 The AI Layer: Gemini 2.5 Flash (Free Tier) Behind a Serverless Proxy, With Computed Tools and Local Verification

Assuming the free-tier key is connected: what is sent, what comes back, and which checks run before an answer reaches the engineer



Output contract: answer text, cited source numbers, confidence 0 to 100, needs_verification, refused. Free-tier quotas apply to the Gemini key; the proxy adds its own per-visitor limits so one user cannot exhaust them. The model can be replaced by any provider that implements the same contract.

Prompt anatomy

System rules p1.1
8 rules: answer from sources, cite [n], never invent values, refuse bypass, treat sources as data

<context>
role, page, equipment, intent, language

<history>
last 6 turns, 500 characters each

<sources> at most 12
id, kind, status, revision, score, text

VERIFIED DRAFT
computed numbers that must stay unchanged

<tool_result> at most 4
2,500 characters each

<question>
user text

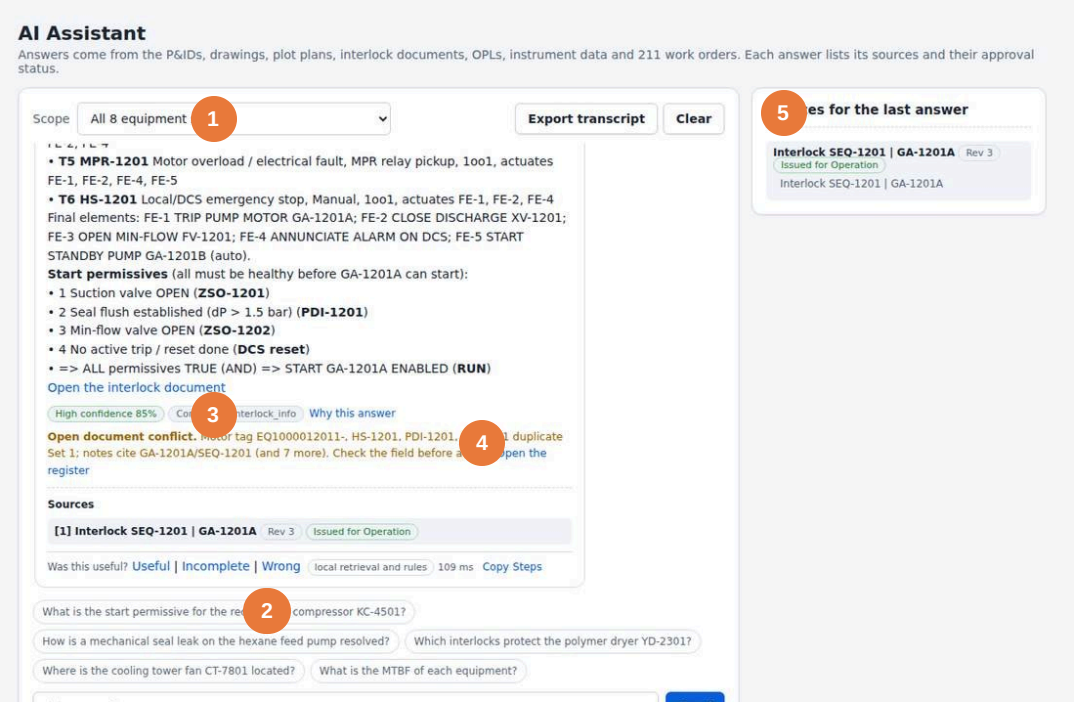
Why this is safe to run on free tier

Key stays on the server. Limits protect the quota. Gemini only words evidence already retrieved or computed, so a failed call costs an answer style, not correctness; the local engine still answers.

Source: Prototype netlify/functions (ai.mjs, extract.mjs, embed.mjs, _lib.mjs) and the page-side assistant (ai.js); prompt version p1.1

A3 The Assistant: Intent Routing, Twenty Computed Tools and an Evidence-Weighted Confidence Score

Counts, rankings and forecasts are calculated from records; Gemini is used to word the result and to answer open questions from retrieved passages



1. Scope selector: the question is limited to one equipment set or the whole plant
2. Suggested questions generated from the data
3. Confidence chip with a breakdown, and the tool that produced the figures
4. Open document conflict note when the answer touches a flagged document
5. Source panel: revision, approval status, record id for every cited item

Twenty tools, grouped

Reliability and cost

maintenance_stats, kpi_summary, failure_modes, predictive_maintenance, failure_memory, history_lookup, spares_outlook

Assets and limits

equipment_info, spec_lookup, tag_facts, plot_plan_query, compare_equipment, rank_equipment

Operations and safety

interlock_info, procedure_help, consequence_analysis, cascade_analysis, hazard_effects, safety_guard

Data governance

data_issues

Intents: chain reaction (utility loss), bypass request, consequence, comparison, ranking, data issues, history, procedure and start-up, diagnosis, spares; anything else goes to retrieval.

Safety guard

A request to bypass or defeat a trip is never answered with steps. The reply names the function that would be defeated (voting, SIL, set point), the consequence from the cause-and-effect matrix, and the approved route: bypass permit, management of change, compensating measures. Covered by safety cases in the evaluation set.

Confidence score, 0 to 100

Question words in one passage	0.45
Match strength of best passage	0.22
Approval status of sources	0.15
Agreement between sources	0.10
Distinct sources found	0.08

Caps after weighting

- Broad question: 60
- Different equipment than asked: 40
- Under half the question words found: 45
- No single passage covers it: 50

High 70+ | Medium 45-69 | Low below 45

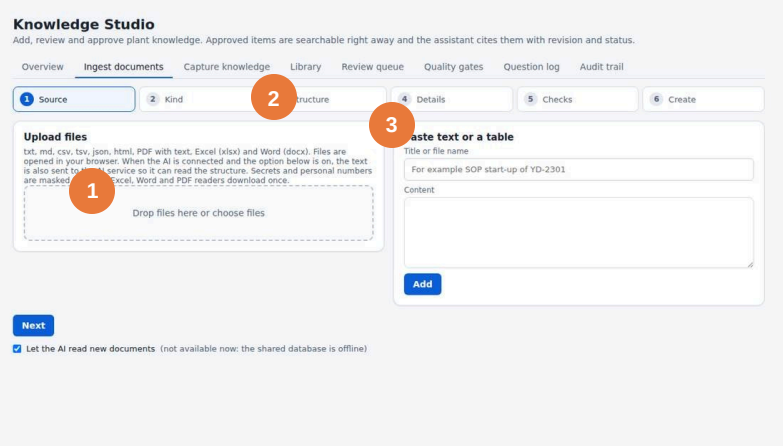
Number and quote check

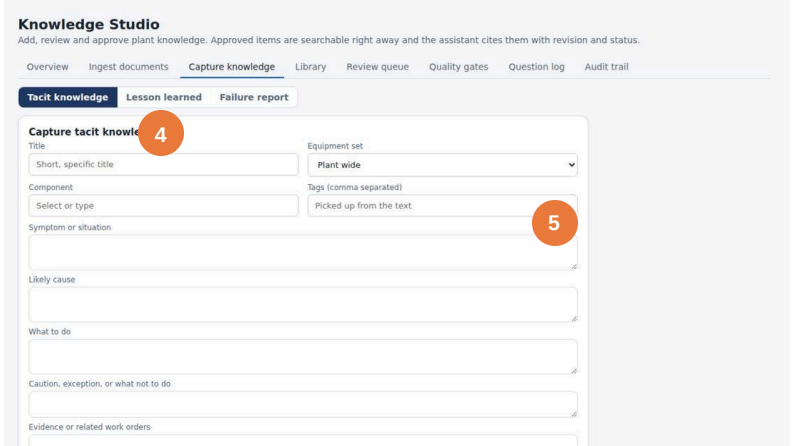
Each figure with a unit (barg, degC, mm/s, m3/h, kW, IDR, %) is looked up in the cited text. Unsupported figures are listed under the answer and the confidence drops. Calculated tools report their own confidence.

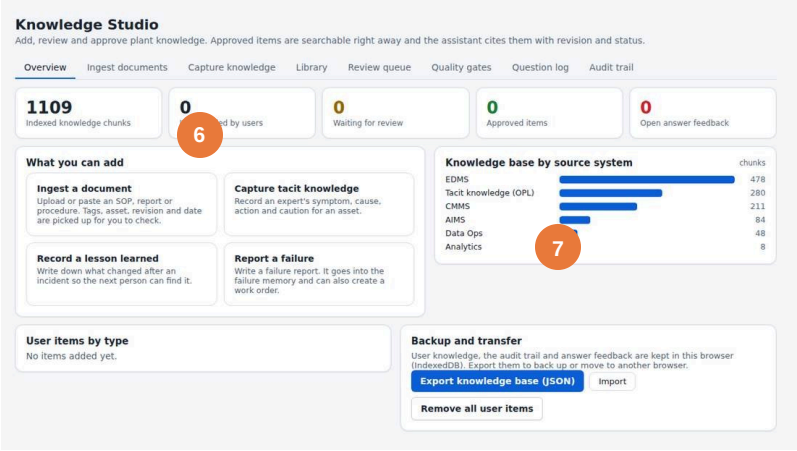
Source: Prototype assistant (ai.js, intent.js, ai2.js); weights and caps are the values used in the running application

A4 Industrial Data Ops in Use: Ingestion Wizard and Knowledge Studio

How a new document, work-order export or piece of tacit knowledge becomes approved, searchable knowledge







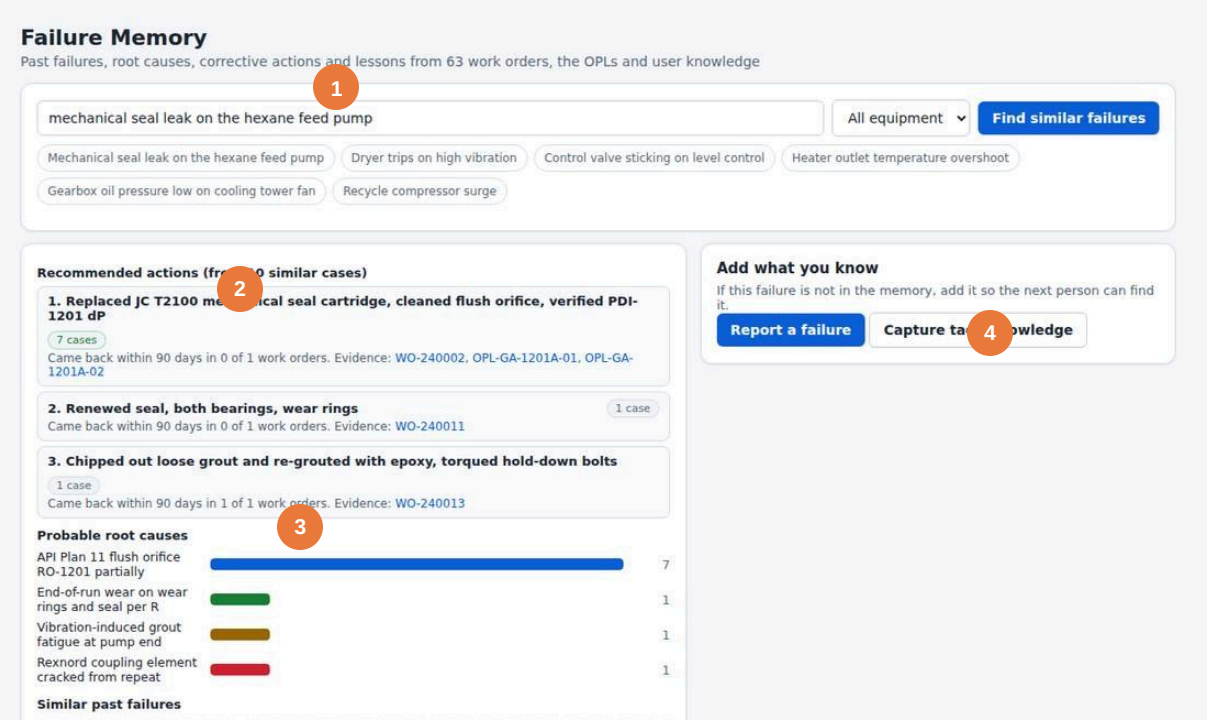
1. Six steps: source, kind, structure or column mapping, details, checks, create
2. Upload txt, md, csv, tsv, json, html, PDF, Excel or Word, or paste text
3. Option to let the AI read the file when connected; secrets are masked first
4. Capture tacit knowledge as symptom, cause, action and caution; failure reports and lessons
5. Each item gets an owner, equipment, revision and a status that starts at Draft
6. Overview, review queue, quality gates, question log and audit trail
7. Reviewers approve or reject; approval makes the item searchable at once

1 Read	2 Verify	3 Map	4 Gate	5 Approve	6 Index
Parsers split CSV or text; with the AI on, /api/extract returns JSON	Each step, value or fact must quote the source word for word or it is dropped	Tags, assets, revision and date detected; columns mapped for tables	Rules: missing owner, unknown tag, conflict with an approved set point	Draft, In review, Approved, Superseded, Rejected; every step audited	Chunks, typed facts and a source copy go to the shared store and the search index

Source: Prototype Knowledge Studio and ingestion wizard (ks.js, wiz.js, ingest.js, facts.js) and /api/extract

A5 Failure Memory: Similar Failures, Root Causes and Actions Retrieved at the Moment of Need

FORESIGHT engine: a symptom is matched against work orders, one point lessons and approved user lessons



1. Free-text symptom, optional equipment filter
2. Recommended actions from the 10 most similar cases, with recurrence within 90 days
3. Probable root causes counted across the matches
4. Add what you know: report a failure or capture tacit knowledge in one click

How a match is built

1	**Retrieve**	Same BM25F index, limited to work orders, OPL cases and approved lessons; top 60
2	**Parse**	Symptom, cause and action fields; component from the part list
3	**Re-rank**	Relevance, plus a boost when the same component is named
4	**Group**	Causes and actions merged by the first four meaningful word stems
5	**Recurrence**	Same component failing again within 90 days
6	**Recommend**	Ranked actions with case counts and downtime range

Where it appears

- Failure Memory page
- Panel inside a work order and the failure view
- failure_memory tool of the assistant
- Scenario and OPL simulator hints

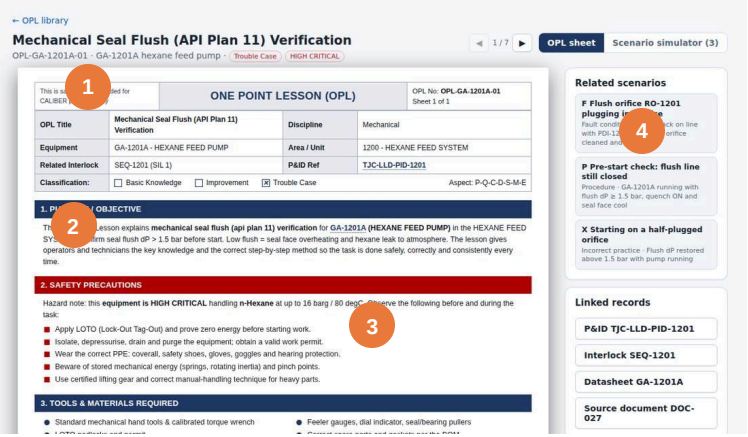
Limits

Matching is lexical with stemming unless embeddings are connected, so it depends on how clearly work orders are written. Each match links to the full record so the engineer judges the fit.

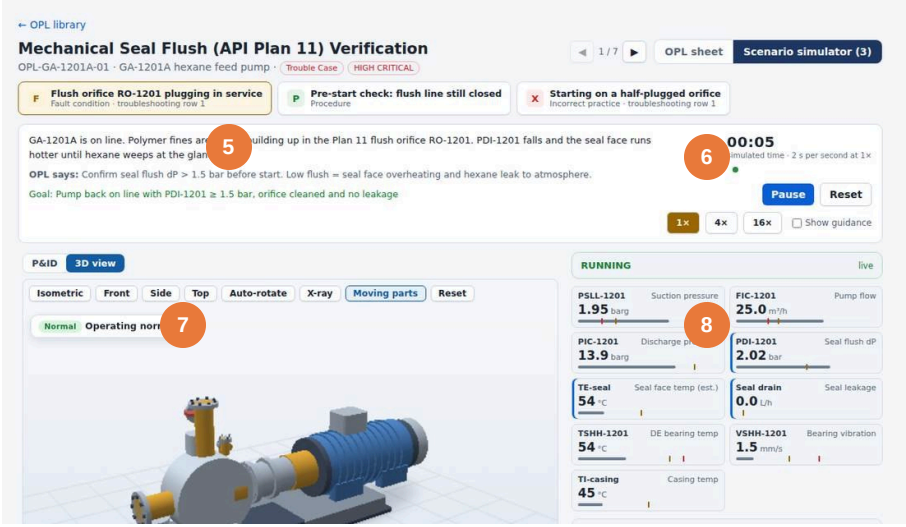
Source: Prototype failure memory (fm.js, pdm2.js) on the case work orders and OPL troubleshooting cases

A6 OPL Library and Scenario Simulator: 56 Lessons You Can Read and Then Play on the Machine

An OPL (one point lesson) is the plant's own short lesson; the simulator turns its troubleshooting cases and steps into a time-stepped exercise



1. Standard OPL header: equipment, discipline, classification, reviewer, approver
2. Safety precautions, tools and materials
3. Detailed steps, each with a check or acceptance criterion
4. Related scenarios, linked P&ID, interlock, datasheet and source document



5. Brief: what the OPL says and the goal; simulated clock with time compression 1x, 4x, 16x
6. 2D P&ID or 3D machine that reacts: shakes, steams, stops
7. Field and panel actions, numbered to the OPL steps; non-standard actions are included on purpose
8. Readouts, trip table, trend, event log and the step checklist

How the simulator works

1. **Process model:** each equipment has variables (pressures, flows, temperatures, levels) integrated in small time steps
2. **Alarm and trip logic:** set points and delays from the interlock document; trips actuate the final elements; a bypass is logged and nothing protects
3. **Operator actions:** valve moves, starts, resets, bypass; the effect shows in the variables
4. **Goal and hold time:** success needs the target state held for a set time; damage ends the run

Score and kinds

Score = 100 - 20 per unsafe action - 12 per trip (up to 3) + 2 per OPL step applied (up to 10); a failed run is capped at 35. Kinds: fault diagnosis, procedure, slow trend, bad practice. The key learning point is shown at the end.

56 OPL documents

280 procedure steps

168 cases

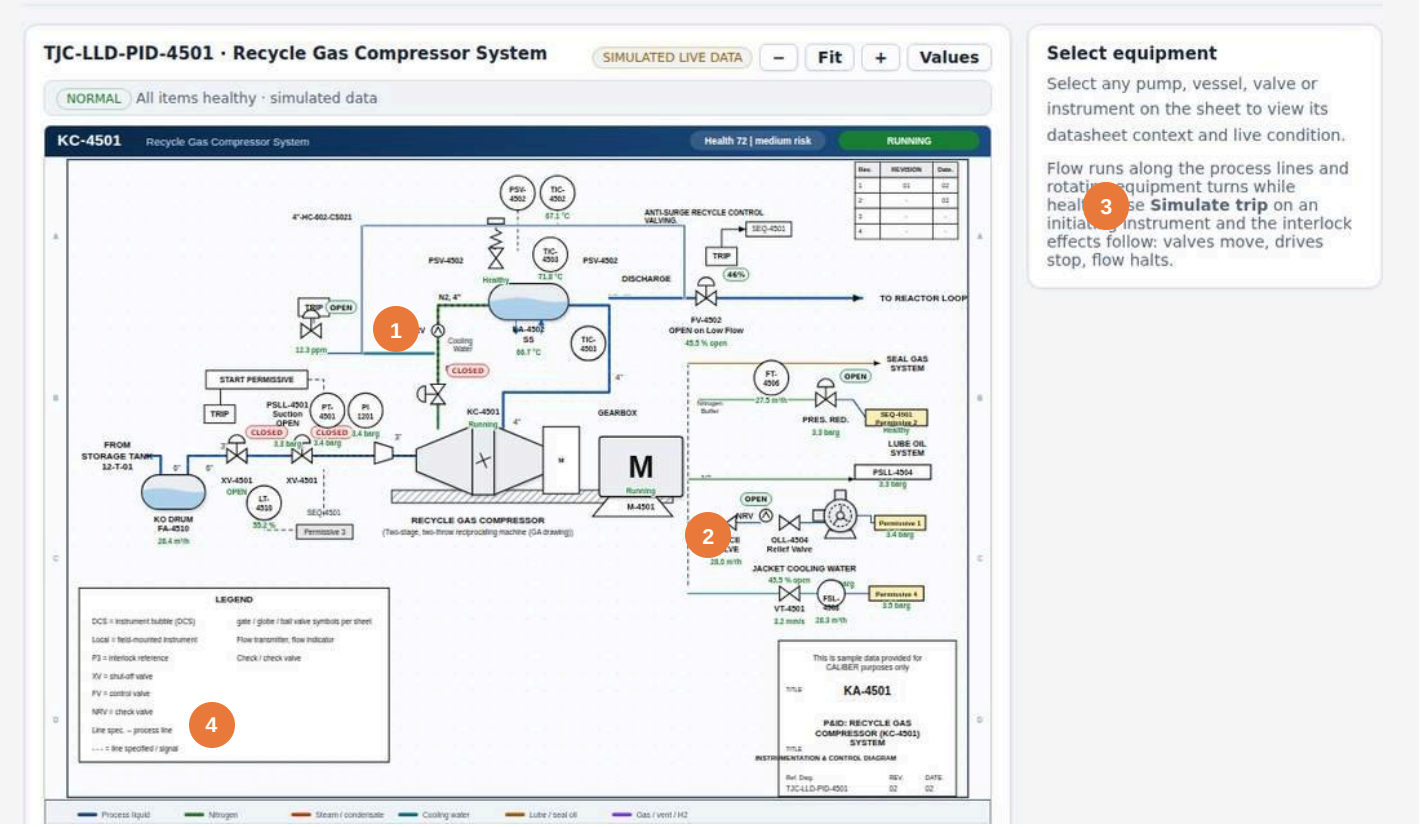
Connected to: Interlock library · Live P&ID · Failure memory · Assistant · 3D machine

The assistant can read any OPL and quote its steps; the simulator reads the same set points, so the lesson and the exercise cannot disagree.

Source: Prototype OPL library and simulator (base.html OPL view and sim engine, sim3d.js); counts from the case OPL documents

A7 Live P&ID: Valves That Open and Close, Flow That Stops and Trips That Follow From the Interlock Logic

Eight P&IDs redrawn as a consistent, service-coloured visual engine and connected to the interlock model, the readiness check and the 3D plant



1. Valve badge: OPEN, CLOSED or percent open; press it, double-click the valve or use the side panel
2. A closed valve stops the flow in the pipe after it and in the pipes that continue from there
3. Instrument bubbles show live values and alarm state; trips latch until reset
4. Service colours: process liquid, nitrogen, steam and condensate, cooling water, lube and seal oil, gas and vent

What the engine does

Flow logic
Pipes are drawn as segments; closing a valve marks every downstream segment as stagnant and stops the flow animation.

Permissives
Closing a valve that proves a start permissive (for example XV-1201 with ZSO-1201) trips the item that depends on it through the same cause-and-effect rows as the simulator.

Automatic trips
Interlock trips close their shutdown valves by themselves, and the badge follows.

Levels
Vessels show a liquid level that moves with the level instrument.

Overview
All eight sheets joined by their battery-limit lines on one plant picture, with a state per unit.

1 Valve	2 Logic	3 Trip
operator closes XV	permissive lost	final elements act

Connected to: Interlock · Readiness · Scenarios · 3D plant

Source: Prototype P&ID engine (pid2.js, pidx.js, pidv.js); 8 sheets, 20 operable valves, 232 tagged items

A8 Reliability Algorithms: Failure-Mode Risk, Bayesian Forecasts and a Backtested Combined Score

FORESIGHT engine: quantitative models built on 211 work orders, validated by rolling-origin backtesting, with their limits stated

1 Failure-mode risk priority

- Mode taxonomy read from problem, cause and action text (10 modes)
- RPN = Severity x Occurrence x Detection
- Severity: percentile of downtime (55%) and cost (45%), scaled 1 to 10
- Occurrence: 1 + 2.4 log2(1 + 3 x rate per year), max 10
- Detection: 1 + 9 x (1 - share found early), minus 1.2 if an OPL covers the mode

2 Gamma-Poisson mode forecast

- Prior from the fleet rate: shape 1.5, rate 1.5 / fleet rate
- Posterior after n own events in T days: shape 1.5 + n, rate prior + T
- P(failure in h days) = 1 - exp(-mean rate x h)
- 80% credible range from the gamma quantile (Wilson-Hilferty)
- Few events shrink toward the fleet instead of over-reacting

3 Measures and trend

- MTBF = (operating hours - downtime) / breakdowns; MTTR = downtime / breakdowns
- Laplace trend test: u above 1.64 means failures accelerating, below -1.64 slowing
- Share of corrective jobs repeating the previous mode
- Preventive-work effectiveness: interval against the usual gap

4 Combined unit score and its validation

- Ridge-regularised logistic regression on eight standardised features: recent breakdowns, time since last breakdown against the usual gap, trend, repeat failures, overdue preventive work, lesson warning signs, recent corrective jobs, low health
- 300 gradient-descent iterations with an L2 penalty
- Rolling-origin backtest: train on the past, forecast the next 90 days, advance 20 days
- Calibration slope shrinks the output toward the base rate

64
forecasts tested

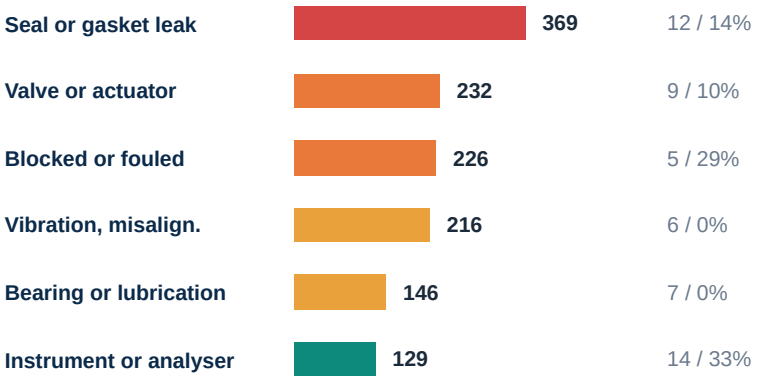
0.26
Brier score, plant-wide rate

0.47
AUC; 0.50 is chance

A unit's own history does not predict which unit fails next on this data. The page ranks units by consequence and reports the combined score as advisory. Improving it needs more history or live condition data, which is a pilot task.

Failure modes by risk priority

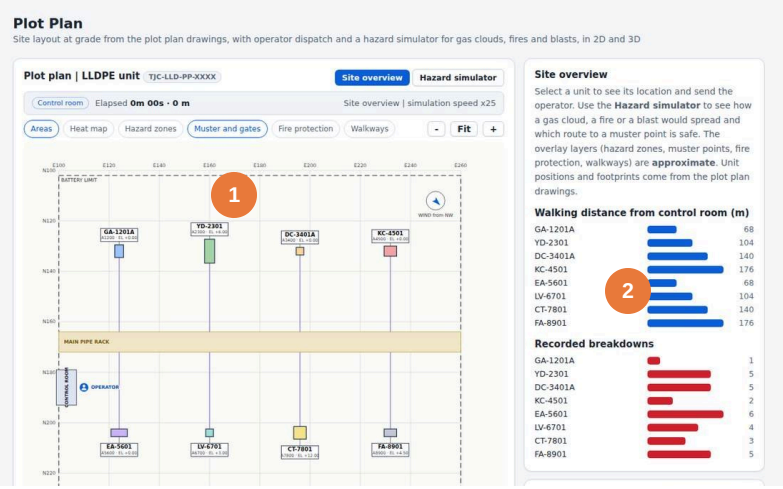
Mode, RPN (bar), jobs / share found early

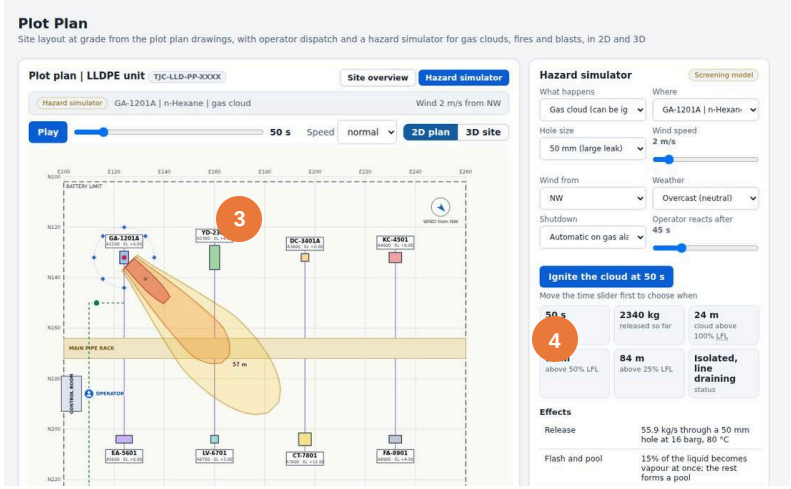


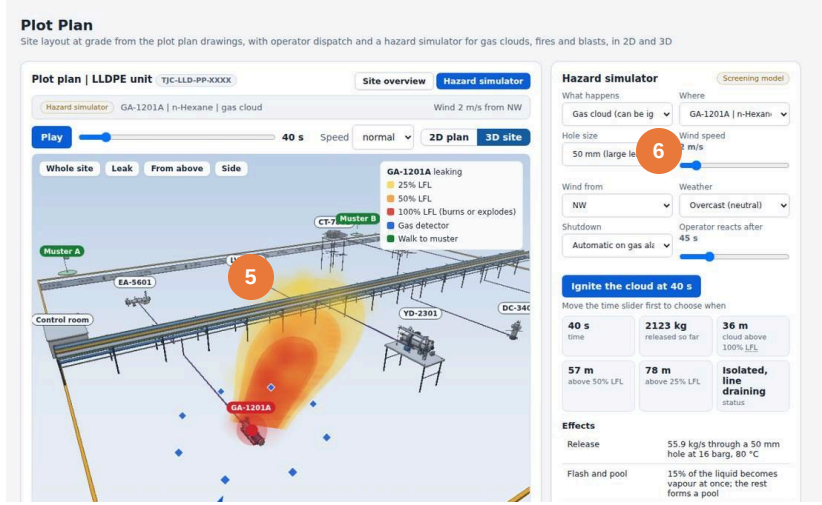
Source: Prototype reliability analytics (pdm.js, pdm2.js); 63 breakdown and corrective jobs behind the failure-mode view

A9 Plot Plan and HSE Hazard Simulator: Release, Cloud, Fire and Muster Walk on the True-Scale Site

Screening models run on the real coordinates of the eight units, the control room, the gates and the muster points







1. Scaled plot plan with E/N grid, pipe rack, road, gates, control room, muster points
2. Colour by downtime, corrective jobs or instruments flagged
3. 2D view: 25, 50, 100% of the lower flammable limit, jet and pool fire, blast zones
4. Controls: fluid, hole 10 to 100 mm, wind, stability class, shutdown mode, ignition time
5. 3D view: gas cloud that grows and drifts, flames, detectors and the muster walk
6. Effects table: release rate, cloud mass, time to detector, time to control room

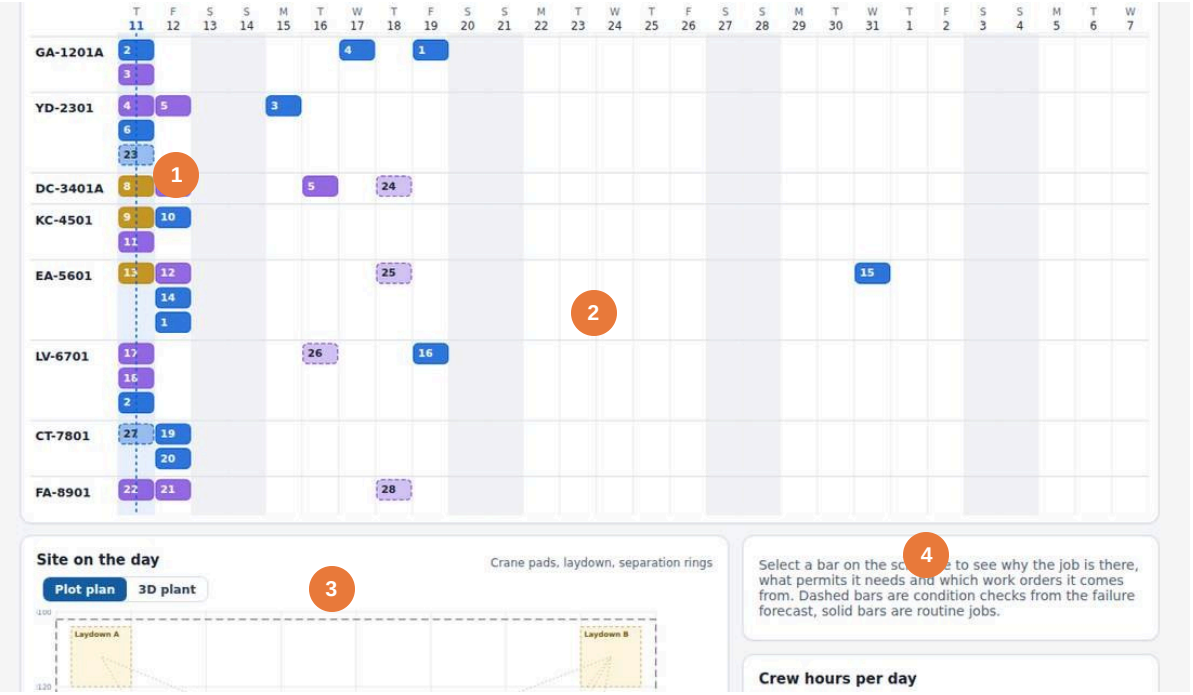
1 Release	2 Source	3 Dispersion	4 Ignition	5 People
orifice flow, hole 10 to 100 mm, shutdown auto, manual or failed	flash fraction and pool evaporation; inventory limits duration	Gaussian plume moving with wind, transport delay, urban Pasquill sigmas, dense-gas factor	flash fire at 25, 50, 100% LFL; jet and pool fire radiation; Kinney-Graham blast	detection and shutdown timing; receptors per zone; walking time to each muster point

Connected to: Site plan | Assistant | Readiness | 3D plant

Source: Prototype plot plan and hazard simulator (pp.js, hse.js, hse3d.js); screening estimates, not a consequence study

A10 Maintenance Site Plan and Plant Replay: Permits, Clashes and the Plant on Any Date

The four-week plan is built from the work-order intervals and the failure forecast; the replay lets a manager watch the plant history on the same map





1. Gantt by unit: solid bars routine jobs, dashed bars forecast checks; letters mark permits
2. Crew hours per day against capacity
3. Site on the day: crane pad, laydown areas, separation rings, permits
4. Clash table with Move to a clear day; add jobs or load an example campaign
5. Date slider, event track and play controls; speed up to very fast
6. 3D plant with each unit's open work order and problem on its label
7. Status board, availability and cost curves up to the chosen date

Jobs	Permits	Schedule	Clashes	Replay
routine: median interval of past jobs per unit, type and discipline; forecast checks from the gamma-Poisson modes (probability 8% or more)	hot work, confined space, lift, isolation, line break inferred from similar work-order wording	greedy by due day, levelled over 3 days, crews per discipline x 8 h; weekends skipped	hot work vs open line at the larger of the setting and 2 x hazard radius; one crane; no hot work with entry	state per unit from work-order times; stepping lands inside every breakdown, so none is skipped

Source: Prototype site planner and plant replay (plan.js, rp.js); figures from the case work orders